

MITHIL K GOWDA

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Professional Summary

Cybersecurity graduate specializing in Artificial Intelligence Security, AI-driven Cyber Defense, Threat Intelligence, Intelligent Security Systems, and Secure Software Engineering. Project Intern at ISRO–LEOS with experience developing secure backend applications using Python, Flask, REST APIs, and MySQL. First author of internationally recognized cybersecurity research focused on multi-agent cyber defense, machine learning for threat detection, adaptive deception, and autonomous security architectures. Seeking graduate opportunities in AI Security Engineering, Intelligent Cyber Defense, Security Research, and Advanced Threat Detection.

Education

M.S. Ramaiah University of Applied Sciences, Bengaluru

2022–2026

B.Tech. in Information Science and Engineering

CGPA: **8.80 / 10**

Technical Skills

AI Security: AI-Driven Cyber Defense, Machine Learning for Cybersecurity, Adversarial AI Fundamentals, Intelligent Threat Detection, Multi-Agent Security Systems, Autonomous Incident Response

Machine Learning: XGBoost, CNN-LSTM, Deep Q-Network (DQN), Feature Engineering, Model Evaluation, Classification Algorithms

Cybersecurity: Threat Intelligence, Threat Detection, Adaptive Deception, Security Monitoring, Threat Modeling, Vulnerability Assessment, Security Architecture, OWASP Top 10

Programming: Python, Java, JavaScript, SQL, HTML, CSS, Bash

Backend Development: Flask, REST APIs, Backend Application Development, API Design

Security Tools: Wireshark, Nmap, Burp Suite, OWASP ZAP

Cryptography: AES-256, SHA-256, Zero-Knowledge Proofs (ZKP), Secure Transaction Verification

Databases: MySQL, MongoDB

Operating Systems: Linux, Windows

Professional Experience

Indian Space Research Organisation (ISRO) – Laboratory for Electro Optics Systems (LEOS), Bengaluru
2026

Project Intern – Secure Software Development & Data Analytics

- Designed and developed secure backend applications using Python, Flask, REST APIs, and MySQL supporting intelligent engineering applications and secure software systems.
- Implemented authentication, authorization, secure API communication, and database security controls emphasizing secure software engineering principles.
- Developed scalable backend services supporting data-driven analytics, secure application workflows, and intelligent system integration.
- Engineered maintainable backend architectures emphasizing software scalability, modular design, secure communication, and enterprise application reliability.

AI Security Projects & Research

A Multi-Agent AI Cyber Defense Framework Using HoneyBee Method

- First Author and Corresponding Author of an AI-driven cybersecurity framework accepted for presentation at ICIICE 2026 (Dubai, UAE) and accepted as a book chapter in the edited volume *Metaheuristic Optimization for Social Good*.
- Designed an intelligent multi-agent cyber defense architecture integrating AI-driven threat detection, adaptive deception, autonomous incident response, and distributed threat intelligence.

- Developed machine learning models integrating XGBoost, CNN-LSTM, Deep Q-Network (DQN), adaptive honeypots, reinforcement learning, and intelligent defensive decision-making.
- Evaluated the framework using CIC-IDS2017 datasets and simulated enterprise attack scenarios, achieving 98.24% detection accuracy, 98.51% precision, 98.10% recall, 0.992 ROC-AUC, and a 1.85% false positive rate.

AI-Governed Carbon Credit Exchange with Digital Carbon Passport

- Abstract accepted at the 4th International Conference on Advancing Sustainable Futures (ICASF 2027), Abu Dhabi University, UAE.
- Designed an intelligent secure platform integrating Digital Carbon Passports, authentication, trust scoring, fraud detection, secure transaction validation, and AI-assisted governance.
- Developed the Secure Trade Authorization and Verification Pipeline (STAVP) utilizing AES-256 encryption, SHA-256 hashing, authentication mechanisms, intelligent transaction validation, and cryptographic security.

C³T-STAVP Cryptographic Framework (Ongoing Research)

- Designing an intelligent cybersecurity framework integrating Character Chaffing & Transposition Technology (C³T), Secure Transaction Authorization & Verification Protocol (STAVP), adaptive trust validation, and AI-assisted security.
- Researching Zero-Knowledge Proofs (ZKP), post-quantum cryptography, semantic obfuscation, intelligent security architectures, privacy-preserving systems, and resilient transaction validation.
- Developing intelligent cryptographic security mechanisms supporting adaptive cyber defense, distributed trust management, and secure enterprise architectures.

Professional Certifications

- IBM IT Fundamentals for Cybersecurity Specialization – IBM
- Systems and Usable Security (Elite) – NPTEL
- Ethical Hacker – Cisco Networking Academy
- Google Cloud Computing – NPTEL
- Introduction to Cyber Security – IGNOU (SWAYAM)
- IBM Front-End Developer Specialization – IBM
- Introduction to Internet of Things (Silver Medal) – NPTEL
- Business Intelligence and Analytics – NPTEL

Achievements

- Selected as Project Intern at the Indian Space Research Organisation (ISRO – LEOS).
- First Author of internationally recognized research focused on AI-driven distributed systems, intelligent software architectures, and cybersecurity.
- Awarded the Reliance Foundation Undergraduate Scholarship (2026).
- Recipient of NPTEL Silver Medal in Introduction to Internet of Things.